

Chanakya University

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System Architecture Design

Prepared the System Architecture Diagram showing the workflow:

User → Chatbot Interface → NLP & Bot Engine → Database → Campus Map (Chanakya University).

Modules and Features

- *User Interface: Designed a chatbot interface for students, staff, and visitors to interact.*
- *NLP & Bot Engine: To process and understand user queries using Rasa/Dialogflow.*
- *Navigation/Pathfinding: Provides campus navigation support using a customised Chanakya University campus map.*
- *Database: Stores information such as campus details, academic departments, staff directory, and FAQs.*

Database Schema Design

- *Designed schema with tables covering:*
- *Locations (academic blocks, classrooms, laboratories, library, auditorium, canteen, hostels, etc.)*
- *Events (seminars, workshops, cultural programmes, and official events at Chanakya University)*
- *Staff Details (faculty members, administrative staff, support staff)*
- *FAQs (frequently asked questions related to admissions, departments, facilities, etc.)*

Methodology

The project follows a systematic methodology to design and implement the Campus Chatbot and Navigation Agent for Chanakya University, Bangalore:

1. Requirement Analysis

Identified user needs such as campus navigation, event information, staff directory, and FAQs.

Defined target users: students, staff, parents, and visitors.

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2. System Architecture Design

Developed a layered architecture consisting of:

User Interface (chatbot front-end)

NLP & Bot Engine (Rasa/Dialogflow for intent recognition and response generation)

Database Layer (campus data storage including locations, events, staff details, FAQs)

Navigation Module (campus map-based pathfinding).

3. Database Design

Created schema with key tables: Locations, Events, Staff, FAQs.

Ensured scalability for adding more campus details in future.

4. Agent Design

Designed the chatbot agent to handle natural language queries from users.

Integrated it with the database to fetch and display relevant results.

Enabled navigation support by mapping queries to campus map locations.

Implementation Approach

Chatbot UI: *Developed using web/mobile interface.*

NLP & Bot Engine: *Configured intents, entities, and training data.*

Backend Integration: *Linked with database and navigation logic.*

Testing: *Verified chatbot responses for accuracy, relevance, and efficiency.*

6. Evaluation

Tested the system with sample queries from students and staff.

Refined the responses and navigation accuracy based on feedback.

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PEAS Description of the Agent

Performance Measure (P):

Accuracy of query understanding.

Correctness of information retrieved (staff details, events, FAQs).

Efficiency in providing navigation within Chanakya University campus.

User satisfaction and response time.

Environment (E):

Chanakya University campus (classrooms, labs, library, administrative offices, hostels, canteen, auditorium).

Students, faculty, visitors, and administrative staff interacting with the chatbot.

Actuators (A):

Chatbot interface (text/voice response to user queries).

Navigation module displaying campus map routes.

Database query engine for retrieving results.

Sensors (S):

User input through chat interface (typed or spoken).

NLP engine interpreting intents and entities.

Database queries capturing campus-related information.